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MINI-REVIEW

Overview on the role of heavy metals tolerance on developing antibiotic resistance in both Gram-negative and Gram-positive bacteria

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Abstract
Environmental health is a critical concern, continuously contaminated by physical and biological components (viz., anthropogenic activity), which adversely affect on biodiversity, ecosystems and human health. Nonetheless, environmental pollution has great impact on microbial communities, especially bacteria, which try to evolve in changing environment. For instance, during the course of adaptation, bacteria easily become resistance to antibiotics and heavy metals. Antibiotic resistance genes are now one of the most vital pollutants, provided as a source of frequent horizontal gene transfer. In this review, the environmental cause of multidrug resistance (MDR) that was supposed to be driven by either heavy metals or combination of environmental factors was essentially reviewed, especially focussed on the correlation between accumulation of heavy metals and development of MDR by bacteria. This kind of correlation was seemed to be non-significant, i.e. paradoxical. Gram-positive bacteria accumulating much of toxic heavy metal (i.e. highly stress tolerance) were unlikely to become MDR, whereas Gram-negative bacteria that often avoid accumulation of toxic heavy metal by efflux pump systems were come out to be more prone to MDR. So far, other than antibiotic contaminant, no such available data strongly support the direct influence of heavy metals in bacterial evolution of MDR; combinations of factors may drive the evolution of antibiotic resistance. Therefore, Gram-positive bacteria are most likely to be an efficient member in treatment of industrial waste water, especially in the removal of heavy metals, perhaps inducing the less chance of antibiotic resistance pollution in the environment.

Keywords Anthropogenic activity · Environmental pollution · Multidrug resistance · Heavy metals

Introduction

The large-scale pollutants resulted from anthropogenic activities have impacts on every part of environment (air–water–soil), drastically influencing the "natural" imbalance. Although the effects of some compounds are still remained arguable; hence, critical scientific reviews/explorations are needed to be addressed. However, microbial life is plentiful in most of the environments: 4000–1,000,000 bacterial species/gram of pristine topsoil (Gans et al. 2005; Zhou et al. 2002; Torsvik et al. 1990). A high microbial diversity can also be sustained either in aerobic or anaerobic condition (Torsvik et al. 1990; Elsgaard et al. 2001; Zhou et al. 2002; Nannipieri et al. 2003; Gans et al. 2005). Therefore, introduction of pollutants in the environment impose significant challenges on the microbial community.

Unlike metabolizable organic pollutants, chemical pollutants (like toxic heavy metals, phenolic compounds) have huge toxic effects on microbial communities, to overcome such stress, some of the bacteria build up cellular defence systems either by expel these toxic compounds or render it less harmful (Nies 1999). Now, resistant or relatively insensitive bacteria are favoured over sensitive types, thereby shifting population sizes of resistant bacteria selected through adaptation by means of selection (Eichner et al. 1999).

The occurrence of genetic changes caused either by mutation or horizontal gene transfer (HGT) play a role in the adaptation and subsequently in selection of bacterial mutants resistance to the polluted environment. Biodegradation/

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bioremediation by such adapted and selected bacteria variants is reproducible and this natural clean-up strategy is indeed sincerely being considered (Röling and van Verseveld 2002). In this notion, an alarming sign has been depicted in various studies that metal contaminations in natural environments may have an important role in emergence and maintenance of antibiotic resistance genes alone with metal resistance in bacteria (Summers et al. 1993; Alonso et al. 2001; Summers 2002). This concern becomes controversial regarding antibiotic resistance pollution, i.e. to maintain a pool of antibiotic-resistance genes in both natural and clinical settings by a long-term selection pressure (co-selection) (Stepanauskas et al. 2005) and it would be a big setback for considering bacteria in bioremediation of heavy metals in waste water treatment plant. In this review, we critically studied phenotypic and genotypic responses (emergence and maintenance of antibiotic resistance by co-selection) of bacteria under toxic heavy stress.

An overview on antibiotic resistance evolution in natural microbial populations

Since discovery of penicillin by Alexander Fleming (1929), antibiotics have become the foundation in the treatment of infectious pathogens. About 160 new antibiotics and semi-synthetic derived compounds were commercialized from 1940s to early 1970s (Gaynes 2017). Despite having great success set up by antibiotics, still bacteria can quickly resist the action of antibiotics and survived, leading to public health crises (Abraham and Chain 1940; Barber 1947; Barber and Rozwadowska-Dowzenko 1948; Smith et al. 2000; Bush et al. 2011; Davies et al. 2013). Although early antibiotic treatments was found to be effective in bacterial infections, but after a while the bacteria became resistant; first erythromycin was commercialised to treat against penicillin-resistant bacteria in 1952 and it became ineffective against 70% strains of *Streptococcus aureus* within 6 months (Finland 1979). 25,000 people die each year due to direct connection with multidrug-resistant bacterial infections (European Centre for Disease Prevention and Control Joint Report with EMEA 2019).

Consistent use of antibiotic imposes a considerable financial challenge on world economies and hence, the rate of antibiotic discovery has been turn down over the past decades: USA expends 35 billion dollars per annum (Livermore 2011; Centers for Disease Control and Prevention 2013). Under such antibiotic crisis, the use of antibiotics has increased globally over the past decades: 63,151 tons antibiotic consumption by livestock in 2010 (Van Boeckel et al. 2015). Now, antibiotics become one of the major pollutants in environments (terrestrial, freshwater and marine environments) due to extensive use of it in the field of aquaculture, foods sector, animal husbandry etc. (Rizzo et al. 2013; Henriksson et al. 2018; Boy-Roura et al. 2018). After all, it is alleged that antibiotic pollution has increased frequency of antibiotic resistance genes in bacteria in different environments, for example, abundance of all kinds of antibiotic resistance genes (ARGs) in soils from the Netherlands was increased radically since 1940s (Knapp et al. 2010). As a result, antibiotic resistant bacteria (ARBs) have now become predominant both in clinics and natural environments. Usually, antibiotic resistance can occur either through mutation by selection or horizontal gene transfer of already existed ARGs in the environment. Mutations in evolution of bacterial resistance led by anthropogenic pressure have now drawn much attention (Gullberg et al. 2011; Kim et al. 2014).

Environmental micro-biomes (barren soil, water column and cave), without influence of any anthropogenic activity or even from pre-dates use of antibiotic contain multiple functional antibiotic resistant genes. Permafrost samples (30,000–5000 year old) could hold resistance genes, although it is evident that resistance genes are radically increased in the present samples than ancient ones (D'Costa et al. 2011; Perron et al. 2015). Therefore, this result articulated that antibiotic resistance genes would more likely to be emerged by means of environmental norms, although strong selection pressures guided by antibiotic, i.e., anthropogenic pressure may also favour the advancement of antibiotic resistance (Bhullar et al. 2012).

Microbial diversity can be shrunken in response to antibiotics exposure; thereby, favouring resistant or tolerant microbial community over susceptible. Moreover, antibiotics increase plenty of parasites and pathogens in soil and water environments, posturing human health at risks (Drury et al. 2013) and also disrupt microbial communities by influencing on bacterial enzymatic activities, such as ureases, dehydrogenases, phosphatises etc. (Cycoń et al. 2019). Generally, antibiotic selective pressures may favour more Gram-negative bacteria than Gram-positive bacteria (Delcour 2009).

Sewage from hospitals is transported to waste water treatment plant (WWTPs) which provides nutrient rich environment to support large bacterial population. Thus, antibiotics at sub-MIC concentrations in this environment may help in the evolution of antibiotics resistance genes by mutation, recombination and dissemination (HGT), (Rizzo et al. 2013). For this reason, hospitals, sewage and WWTPs are described as hotspots for the evolution of resistant bacteria. At the end, WWTPs may act as buffer zone of interference to restrain resistance evolution and therefore, during bioremediation, accurate precision must be taken in consideration of environmental ecology on bacterial evolution.

Effect of heavy metals on microbial ecology

Heavy metals, such as Cr, Ni, Cu, Zn, Cd, Pb and Hg are well-known environmentally relevant most hazardous pollutants due to its toxic, persistent and bio-accumulative nature (Türkmen et al. 2009; Jović et al. 2012; Ramírez 2013; Rahim et al. 2016). Weathering of rocks and volcanic eruptions, mining, fossil fuels combustion, various industrials emissions, agricultural activities (pesticides and phosphate fertilizers) etc., are considered to be the natural sources of heavy metals in environment. It has a significant influence on the environment by contaminating food chains and induces different health problems, i.e. serious diseases or even death of organisms due to their toxicity (Wieczorek-Dąbrowska et al. 2013). Heavy metals, such as Mn, Fe, Co, Ni, Cu, Zn, Mo and As (metalloid) are beneficial micronutrients or trace elements for organisms (Li et al. 2008; Waldron et al. 2009; Schlesinger et al. 2011). Bacteria are ubiquitous in nature and are associated with major biological sphere of life. Microbial communities in soil perform many functions like nutrient cycling, symbiosis with plant and detoxification of chemicals. Thus, microbial populations may act as useful performer for improvement of soil quality (Zhou 2003; Verma and Singh 2005; Ivshina et al. 2014). Generally, heavy metals in soil change microbial community structure by reducing microbial biomass and functional diversity (Frostegård et al. 1993; Gadd 1993; Sandaa et al. 2001; Akmal et al. 2005). On the contrary, high metal concentrations may also develop metal resistant or tolerant microbial populations (Ellis et al. 2003). Dynamics of microbial communities in polluted environments are quite significant: highly contaminated sites are likely to favour less bacterial diversity, eventually this scenario leads to build up a reduced gene variation with plenty of specific functional genes, viz., heavy metal resistance genes and sulphate-reducing genes (Waldron et al. 2009; Kang et al. 2013). On other hand, abundance of species diversity was observed in the moderately contaminated sites, consisting of the leading groups, *α-Proteobacteria*, *β-Proteobacteria* and *Firmicutes* (Zhu et al. 2013). Microbial community structure was appeared to be shifted under the gradient of stress conditions, indicating that the diversity of microorganisms is highly influenced by heavy metals.

Bioremediation of heavy metals by microorganism

The heavy metal pollutants from their sources are released into the environments (air–water–soil) and effects on human health and biological organisms by high accumulation in the natural system (Dixit 2015). Thus, it becomes compulsory to control hazardous heavy metal pollutions in the environment. Industrial effluents contaminated with heavy metals are usually treated by two well-known methods, chemical and biological (Ahluwalia and Goyal 2007; Kuan et al. 2009; Mouedhen et al. 2009). Chemical approaches are expensive and have several disadvantages, such as incomplete metal removal, requirement of huge reagent and energy and generation of toxic sludge. Biological approach has the great potential that can be economical for achievement of the goal.

Many microorganisms have the unique ability to develop resistance mechanisms and can detoxify/remove the heavy metals (Monachese et al. 2012; Sharma 2012). Consequently, bioremediation of heavy metals in polluted soils and industrial effluent treatment plant using microorganisms is regarded as a potential alternative over chemical methods for environmental restoration (Letters 2011). It employs bacteria, fungi and plants for purification (detoxification/accumulation/removal) of different environmental pollutants (Dua et al. 2002). In the recent past, it is quite predictable that bioremediation using microorganisms is low cost and eco-friendly (Rajendran et al. 2003). In addition, unlike other pollutants, heavy metals cannot be degraded by microbial bioremediation, rather accumulation and detoxification (i.e. to make in less toxic form) can only be done by interaction of microbe with heavy metals (Rajendran et al. 2003). Bacterial strains adopt different mechanism like heavy metal tolerance or resistance, viz., transportation across cell membrane, accumulation on cell wall, biosorption, formation of complexes, reduction or oxidation and intra or extracellular precipitation (Veglio et al.1997; Ahemad 2012; Hrynkiewicz and Baum 2014). Among them, biosorptive processes are more appropriate than the bioaccumulative processes. Biosorption is proven to be low cost and environment friendly for removing toxic metal ions, as because it is passive and metabolic independent (Volesky 1990). Although bioaccumulative processes involve metabolic systems, i.e. active uptake often requires addition of nutrients. Consequently, it increases biological oxygen demand/chemical oxygen demand in the environment (Hussein et al. 2004).

Most of the resistant determinant genes in bacteria are located on plasmids (e.g. Sb^{3+}, Zn^{2+} resistance), but very few determinants are also identified in chromosome-like *Pseudomonas* sp. (KR23), *E. coli* (KR29) (Cavicchioli and Thomas 2002; Jasmine et al. 2012; Chihomvu et al. 2015). So, resistance genes may either be located on chromosome or plasmids of bacteria (Ji and Silver 1992; Diorio et al.1995). During the mechanism of tolerance, bacteria may develop co-selection for antibiotic resistance that has led to a new area argument and interests. Later section of the discussion is expected to address the better sense of correlation between antibiotic resistance development and heavy metal tolerance by bacteria.

Response to heavy metals by both Gram-positive and Gram-negative bacteria

Heavy metal resistant bacteria are usually engaged for the bioremediation of heavy metal. At first, the heavy metals cations are get trapped and then accumulated on bacterial cell wall through electrostatic interactions with anionic nature of bacterial surface that is constituted by different functional groups, carboxyl and amine groups, etc. (Doyle et al. 1980; Kang et al. 2007). This approach is metabolism independent (less cost effective), i.e. it is driven by without any ATP investment. Primarily, peptidoglycan layer from both Gram-positive and Gram-negative bacteria determines anionic nature of cell surface showing the metal binding ability. In addition to peptidoglycan with widen thickness, cell wall of Gram-positive bacteria contains teichoic acid and teichuronic acids, which provide more anionic character for high-metal binding capacity (Beveridge and Murray 1980). Therefore, Gram-positive appear to accumulate higher concentrations of heavy metals on their cell walls as compared to Gram-negative bacteria (Rani and Goel 2009).

In Gram-negative bacteria (e.g. *Ralstonia eutropha*, *Pseudomonas aeruginosa*), heavy metal resistance gene *czc* was identified as metal active efflux system (CzcABC) which is responsible for the resistance to Cd, Zn and Co (Nies 1995). Three cadmium resistance gene clusters, *cadA2R*, *czcCBA1* and *colRS* were recognized in *Pseudomonas* sp., where *cadA2R*, *czcCBA1* were meant for efflux systems (Nowicki et al. 2015). ncc system in *Alcaligenes xylosoxidans* operates the similar mechanism for resistance to Ni, Cd and Co. *Pseudomonas syringae pv. tomato* confiscates Cu in periplasm and outer membrane by Cu-binding proteins (*cop* genes) and *E. coli* develop Cu resistance mechanism by ion-dependent Cu antiporter system (*pco* gene) (Kunito et al. 1998). Gram-positive bacteria (e.g. *Staphylococcus*, *Bacillus* or *Listeria*) usually resist Cd by efflux system, Cd-efflux ATPase. The efflux pump systems were found to be effective approach, mostly in Gram-negative bacteria, where efflux pump systems instantly modulate metal toxicity. On the other hand, binding and storage/accumulation of heavy metals may induce long-term stress tolerance, as it is portrayed in Gram-positive bacteria.

Relevance of co-selection in the evolution of resistance to heavy metals and antibiotics

Co-selection mechanisms relate to co-resistance (resistance mechanisms to metals and antibiotics are present on the same genetic element) and cross-resistance (the same mechanism responsible for simultaneous resistance to metals and antibiotics). Mercury resistance gene *merA* was studied extensively in bacteria like members of *Enterobacteriaceae*, *Pseudomonas*, *E. coli* etc., and in most of the case, it closely linked to various combination of antibiotic resistant genes (integrons) (Wireman et al. 1997; Bass et al. 1999). *merA* was supposed to determine multidrug resistance by co-resistance. In subsequent study, isolates of Enterobacteriaceae with genetic determinant *merA* in heavy metal contaminated site (As, Cr, Cu, Hg,) did not develop multidrug resistance phenotype (Henriques et al. 2016). On the contrary, less concentrated heavy metal contaminated sites harbouring isolates, including *Shewanella*, *Vibrio*, *Pseudomonas* and members of Enterobacteriaceae without having any genetic determinant showed significant resistance phenotype. Both the isolates, members of *Enterobacteriaceae* and *Pseudomonas* with or without *marA* from highly contaminated sites exhibited almost similar pattern of resistance phenotype (Henriques et al. 2016). In another study, it was reported that copper resistance gene *tcrB* was found to be physically linked to *ermB* gene (macrolide resistance) and Tn1546 element (glycopeptide resistance) in *Enterococcus faecium*, although copper (copper sulphate) exposure alone was not enough to continue resistance to macrolide and glycopeptides; while resistance to these antibiotics was declined after banning their use in growth promotion of pigs (Hasman et al. 2005).

Co-regulatory resistance (transcriptionally linked regulatory systems) is considered as one of the mechanism of co-selection. For instance, antibiotic resistance gene *oprD* (imipenem resistant determinant) and a heavy metal efflux pump encoding gene, *CzcCBA* are transcriptionally linked (Conejo et al. 2003; Perron et al. 2004). Upon exposure of toxic Zn, the isolates of *Pseudomonas aeruginosa* became resistant to heavy and antibiotic imipenem by activating CzcCBA and deactivating oprD. This phenomenon is reversible, where OprD expression is very much modulated by environmental conditions, such as carbon or nitrogen in the environment (Conejo et al. 2003). In *E. coli*, *mdtABC* operons (formerly *yegMNO)* encode an RND-type efflux system, which confers resistance to novobiocin, deoxycholate and bile salts. *mdtABC* expression is up regulated by overproduction of the response regulator/down regulator BaeR under excess of Zn concentration. Majority of genes in *mdtABC* operons become reduced by decreasing induction ratios for the downstream genes *BaeR*, because Zn is an essential trace element for growth (Lee et al. 2005). Hence, this way of occurrence of antibiotic resistance in the bacteria was seem to be temporary and may not contribute antibiotic resistance pollution in the environment.

Burkholderia cepacia became susceptible to both heavy metals (Cd^{2+} and Zn^{2+}) and antibiotics, including beta-lactams, kanamycin, erythromycin, novobiocin, ofloxacin and sodium dodecyl sulphate, by mutation in DsbA–DsbB system (Hayashi et al. 2000). Hence, in *Burkholderia cepacia,* wild form of DsbA–DsbB system may function as cross-resistance.

Therefore, based on the quality of available data, co-resistance, co-regulation and cross-resistance induced by heavy metals stress were unlikely to develop MDR in environmental bacteria (Fig. 1).

Reality check on impact heavy metal resistance in bacteria on antibiotic resistance

Increasing antibiotic resistance genes (ARGs) and antibiotic resistant bacteria (ARB), i.e. antibiotic resistance pollution and its rapid dissemination in hospital settings create a significant threat to public health. Gram-negative bacteria are of particular concern about rising in hospital-acquired infections. Under selective pressure, Gram-negative can develop resistance by acquiring diverse range of antibiotic resistance genes in the natural environment and clinic (Mauldin et al. 2010; Ruppe et al. 2015; Nelson et al. 2017). Generally, Gram-negative are easily trend to develop multidrug resistance compared to Gram-positive bacteria (Table S1) (Delcour 2009; Rossolini et al. 2014; Hawkey 2015). The environmental factors are steadily recognized as an important force for emergence and spread of antibiotic resistance genes (Baquero et al. 2008; Finley et al. 2013). Other than antibiotics, various factors in the environment may crucially play an important role in evolution of antibiotic resistant bacteria (Gaze et al. 2005; Tacão et al. 2012). Heavy metal pollutants are supposed to be an alternative force of antibiotic and it may act as a co-selection pressure of antibiotic resistance (Bednorz et al. 2013; Stepanauskas et al. 2006). In this context, unlike Gram-negative bacteria, Gram-positive bacteria are highly stress tolerant as because it accumulates high amount of heavy metal, whereas heavy metal tolerance in Gram-negative bacteria are mostly operated through efflux pump, resulting avoidance of prolongs stress tolerance (Nanda et al. 2019). Therefore, development of multidrug resistance both in Gram-negative and Gram-positive bacteria are not directly correlated or influenced by toxic heavy metal tolerance (Table S1). Let's put the fact into perspectives and try to understand the relationship between antibiotic resistance and heavy metal resistance or tolerance both in Gram-negative and Gram-positive bacteria. *Staphylococcus hominis* strain AMB-2 was highly tolerant to multiple heavy metals and showed susceptibility to available sixteen antibiotics (Zeeshanur et al. 2019). Total nineteen bacterial strains (DU1–DU19) of *Enterobacter sp.* resistant to Cr (VI) were isolated from tannery waste dump site. Out of it, strain DU17 selected for efficient Cr (VI) reduction showed very low multidrug resistance (Rahman et al. 2014). Only ten strains reduced Cr (VI), suggesting resistance or tolerance did not correlate to its removal approach (Viti et al. 2003; Polti et al. 2007). Multidrug-resistant *E. coli* (six strains) and *Salmonella* spp. (six strains) were isolated from raw salad vegetables (RSVs), which contain permissible limit of heavy metals (Ahmed et al., 2019). Multiple strains of six

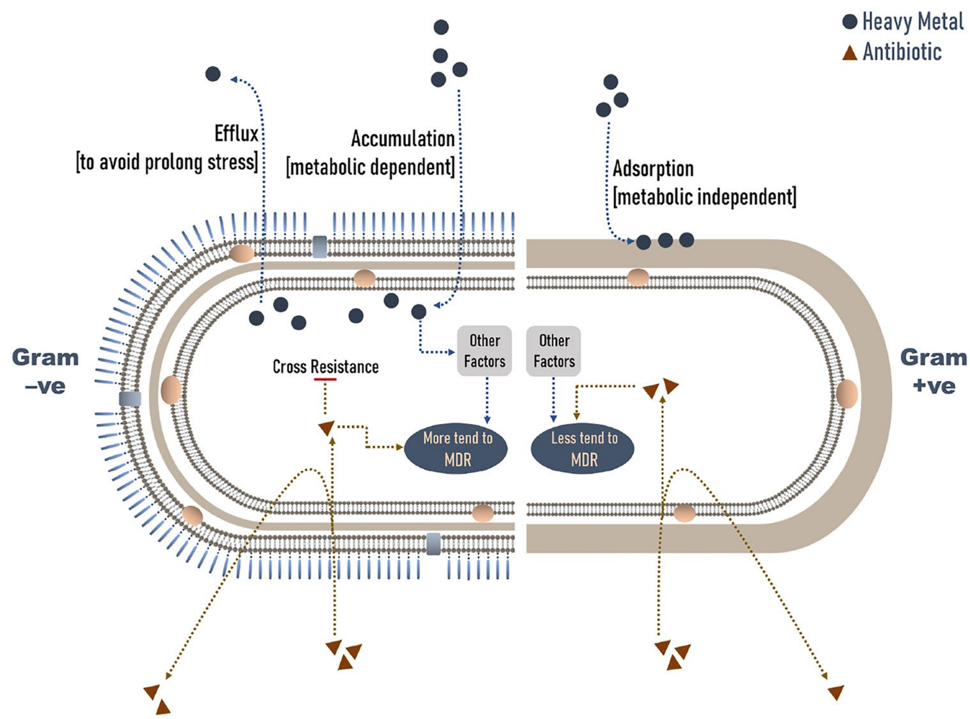

Fig. 1 Schematic representation of assumed hypothesis: Gram-positive bacteria showing less proneness to become MDR are highly heavy metal tolerance, whereas Gram-negative bacteria responded as opposite to Gram positive, i.e. more prone of being MDR and also showing the avoidance of heavy metals tolerance by efflux pump systems. These phenomenons may result a conclusion that no one single source of factors (e.g. heavy metals) in the environment would likely to induce MDR by co-selection; combination of environmental factors and genotype of bacteria may favour the advancement of MDR

Gram-negative bacteria, *Shewanella*, *Comamonas*, *Vibrio*, *Enterobacteriaceae*, *Pseudomonas* and *Aeromonas* were isolated from three sites of salt marshes located in Ria de Aveiro (heavy metal concentrations in two highly contaminated sites were threefold higher than control site), a significantly inconsistent correlation (heavy metal resistance *vs*. antibiotic resistance) was observed between lesser contaminated control sites with higher contaminated sites, with many exceptions (Henriques et al. 2016) (Table S1). Based on the recorded data in the supplementary table, all these observations have been represented and decorated in Fig. 2.

Conclusion

Continued evolution of ARGs is emerging threats to public health and has challenged to frame strategies for antibiotic resistance pollution management. Antibiotic resistance genes are largely being disseminated among the pathogenic bacteria in hospital, community and the natural environment. Rapid demographic, environmental, industrial and agricultural changes may add a global crisis for the evolution of multidrug resistance. If this is continued to prevail that is going to cause many death in the next coming decades.

Strategy-based approaches are expected to improve our understanding about the environmental risks posed by hazardous material including ARGs. Effective measure should be taken for bioremediation of toxic heavy metals. Still,

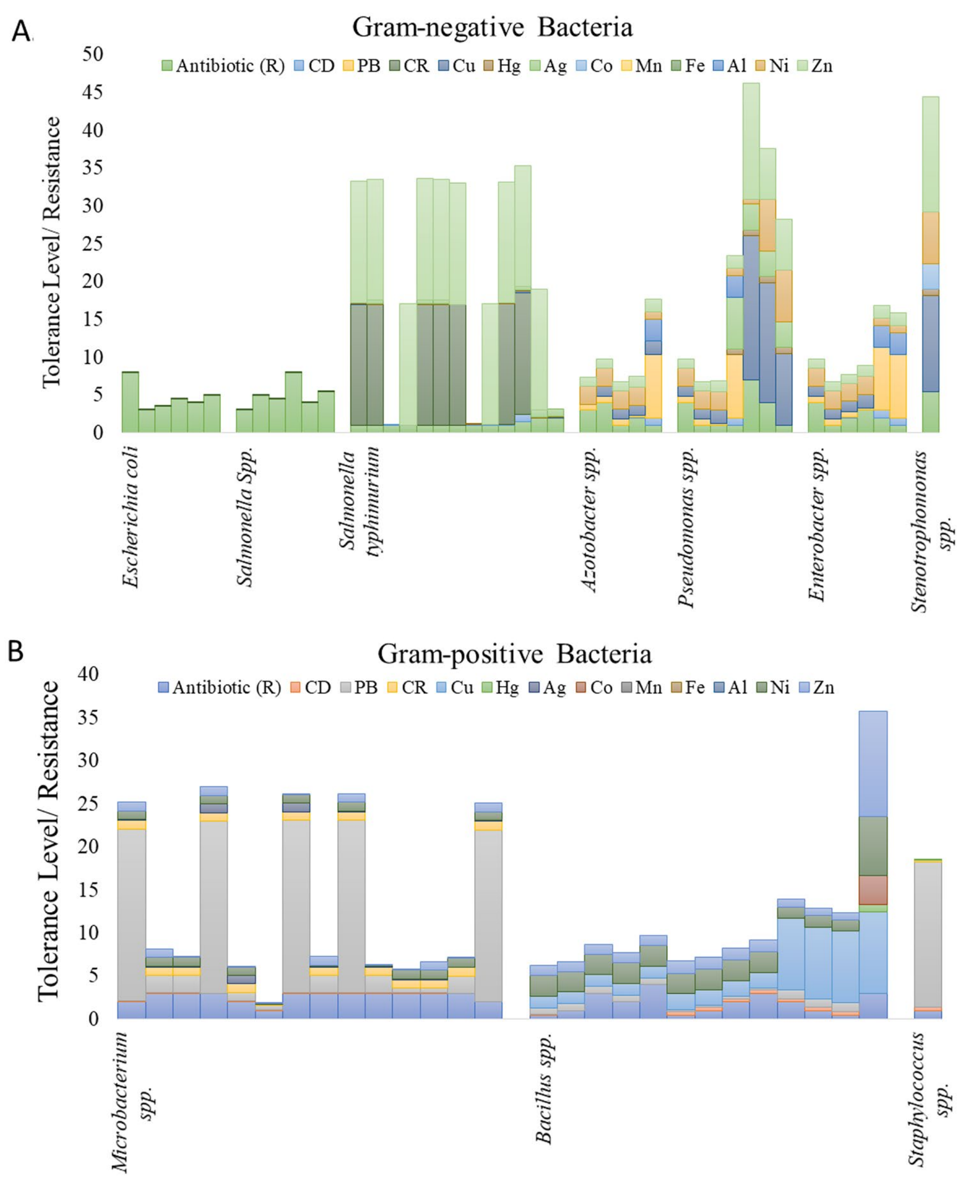

Fig. 2 Both the Gram-negative and Gram-positive bacteria show their resistance development to both antibiotics and heavy metals, which are represented by number of different antibiotic resistance (R) and metal tolerance in milli-Molar (mM) concentration, respectively. **a** Gram-negative bacteria consistently develop antibiotic resistance response without any significant quantity of heavy metal exposure or vice versa. **b** Gram-positive bacteria showed their adaptability against different types of heavy metals and these bacteria are seemed to be less prone of being multi-drug resistance and these bacteria may have less risk for developing antibiotic resistance pollution in the environment and are appeared as potential metal tolerant agent during bioremediation

many observations create controversy regarding the contribution of toxic heavy metals on developing antibiotic resistance. Hence, many more researches are required to resolve the emerging issue. Gram-positive bacteria are highly stress tolerant as because it accumulates more heavy metals and are unlikely to become MDR compared Gram-negative bacteria. While, selective pressure is likely to favour the advancement of MDR in Gram-negative bacteria. Therefore, Gram-positive bacteria in the environment may have a convincing role in bioremediation of toxic heavy metals. However, there was paradox of heavy metal tolerance or resistance with developing MDR in the bacteria. Other than antibiotics, a combination of environmental factors may influence in the evolution of antibiotic resistance.

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Author contributions RB 1 and 2 adopted the idea. RB 1, UH, AK and AM collected information. RB 1, 2 wrote and all authors read and approved the final manuscript.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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